MAHATMA GANDHI UNIVERSITY, KOTTAYAM
Answer Key
MGU-BVOC (HONOURS) – HUMAN PHYSIOLOGY
MG1SDCSNP102

(2025 Admission Onwards)

Duration: 1 hour 30 minutes **Max. Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I), Appreciation (Ap)

Part A
Short Answer Questions
(Answer any 10 questions)

1. Define diffusion and osmosis.

Answer: Diffusion: Movement of molecules from high to low concentration.
Osmosis: Movement of water across a semipermeable membrane toward higher solute concentration. (CO1, K)

2. What is homeostasis? Mention two examples.

Answer: Homeostasis is the maintenance of internal stability. Examples: thermoregulation, blood glucose regulation. (CO1, K)

3. Mention the types of ECF.

Answer: Types of ECF: Interstitial fluid, plasma, transcellular fluid. (CO1, K)

4. Enumerate the sites of erythropoiesis.

Answer: Sites: Red bone marrow of long bones, sternum, ribs, pelvis, vertebrae. (CO2, K)

5. Compare the functions of neutrophils and eosinophils.

Answer: Neutrophils: Phagocytosis.
Eosinophils: Defense against parasites & allergic reactions. (CO2, An)

6. Mention the structures involved in mechanism of muscle contraction.

Answer: Actin, myosin, tropomyosin, troponin, sarcoplasmic reticulum. (CO2, K)

7. Define coagulation time and state its normal value.

Answer: Coagulation time = time for blood to clot; Normal: 3–8 minutes. (CO2, K)

8. What happens during ventricular systole?

Answer: Ventricles contract, blood ejected into aorta & pulmonary artery; AV valves close, semilunar valves open. (CO3, K)

9. Define systolic and diastolic blood pressure.

Answer: Systolic: Pressure during contraction.
Diastolic: Pressure during relaxation. (CO3, K)

10. Define vital capacity and its normal value.

Answer: Vital capacity = Max air exhaled after maximum inhalation; Normal: 3–5 L. (CO3, K)

11. Mention the cranial nerves involved in taste sensation.

Answer: Cranial nerves: Facial (VII), Glossopharyngeal (IX). (CO4, K)

12. What is anemia?

Answer: Anemia is decreased hemoglobin or RBC count. (CO2, K)

Part B
Short Essay Questions
(Answer any 6 questions)

13. Draw a labelled diagram of a cell.

Answer: Diagram to be drawn separately (include nucleus, mitochondria, ER, Golgi, cell membrane). (CO1, A)

14. Differentiate procoagulants and anticoagulants.

Answer: Procoagulants: Promote clotting (e.g., fibrinogen). Anticoagulants: Prevent clotting (e.g., heparin). (CO2, An)

15. Classify neurotransmitters with examples.

Answer: Excitatory: ACh, glutamate; Inhibitory: GABA, glycine; Monoamines: serotonin, dopamine. (CO2, An)

16. Explain the digestion in the stomach.

Answer: Stomach secretes HCl, pepsin; proteins begin digestion; mechanical churning occurs. (CO3, U)

17. Describe the segments and uses of ECG.

Answer: P wave, QRS, T wave; ECG helps detect arrhythmias, ischemia, conduction defects. (CO3, U)

18. What are the defects of visual field?

Answer: Defects include hemianopia, scotoma, tunnel vision. (CO4, U)

19. Explain the role of kidney in water balance.

Answer: Kidney uses ADH, aldosterone, tubular reabsorption to maintain water balance. (CO4, U)

20. Explain the process of urine formation.

Answer: Steps: Filtration, reabsorption, secretion. (CO4, K)

21. Write a note on feedback control systems involved in thermoregulation.

Answer: Thermoregulation via hypothalamus using negative feedback: sweating, vasodilation, shivering. (CO1, U)

Part C
Essay Questions
(Answer any 2 questions)

22. Describe in detail the composition and functions of blood.

Answer: Blood composition: Plasma + formed elements (RBCs, WBCs, platelets). Functions: transport, immunity, temperature regulation, pH balance. (CO2, U)

23. Explain the anatomical structure and functions of the human heart.

Answer: Heart: Four chambers, valves, conduction system; Functions: pumping blood, maintaining circulation. (CO3, U)

24. Explain spermatogenesis and factors affecting spermatogenesis.

Answer: Spermatogenesis: Spermatogonia → mitosis → meiosis → spermiogenesis. Factors: hormones, temperature, nutrition. (CO4, U)

25. Explain the regulation of fluid balance and the clinical conditions of dehydration and edema.

Answer: Regulation via ADH, aldosterone, thirst mechanism.
Dehydration: reduced body water.
Edema: excess interstitial fluid. (CO1, U)